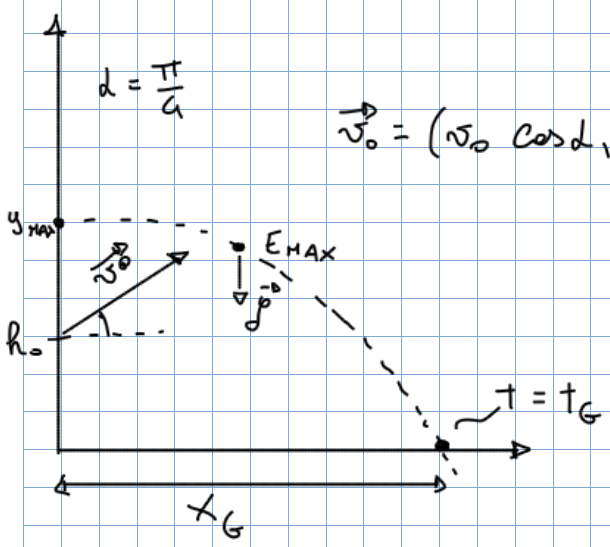
Exercício 15/16



$\vec{v_0} = (v_0 \cos\alpha, v_0 \sin\alpha)$

$E_{MAX} = ?$

$x_G = ?$

$y_{MAX} = ?$

$x(t) = (v_0 \cos\alpha)\, t$

$y(t) = h_0 + (v_0 \sin\alpha)\, t - \frac{1}{2} g t^2$

$v_y(t) = v_0 \sin\alpha - gt \quad \left(= \frac{dy}{dt}\right)$

$v_y(t_{MAX}) = 0 = v_0 \sin\alpha - g t_{MAX} \iff t_{MAX} = \frac{v_0 \sin\alpha}{g}$

$y(t) = 0 \quad$ com $t = t_G \iff y(t_G) = 0, \; x = x_G$

$\Downarrow$

$h_0 + (v_0 \sin\alpha)\, t_g - \frac{1}{2} g t_G^2$

$\frac{1}{2} t_g^2 - \frac{v_0 \sin\alpha}{g} t_g - \frac{h_0}{g}$

$a = \frac{1}{2} \quad b = -\frac{v_0 \sin\alpha}{g} \quad c = \frac{h_0}{g}$

$b^2 - 4ac = \left(\frac{v_0 \sin\alpha}{g}\right)^2 + \frac{2h_0}{g} = \frac{b^2 - 4ac}{4a^2}$

$t_g = \frac{v_0 \sin\alpha}{g} \pm \sqrt{\left(\frac{v_0 \sin\alpha}{g}\right)^2 + \frac{2h_0}{g}} \Rightarrow t_G = \frac{v_0 \sin\alpha}{g} + \sqrt{\left(\frac{v_0 \sin\alpha}{g}\right)^2 + \frac{2h_0}{g}}$

$x(t=t_G) = (v_0 \cos\alpha)\, t_G$

$h_0 = 1\,m$

$\alpha = \frac{\pi}{4} = 45°$ $\Rightarrow t_{MAX} = 0.36$

$v_0 = 5\,m/s$ $t_G = 0,94$

$g = 9.81\,m/s^2$ $x_G = 3,31\,m$

$0 = v_y(t_{MAX}) = v_0 \sin\alpha - g\, t_{MAX} \Rightarrow t_{MAX} = \frac{v_0 \sin\alpha}{g}$

$y_{MAX} = y(t_{MAX}) = h_0 + \frac{(v_0 \sin\alpha)^2}{g} - \frac{1}{2} g \frac{(v_0 \sin\alpha)^2}{g^2} = h_0 + \frac{(v_0 \sin\alpha)^2}{-g} = 1,64\,m$

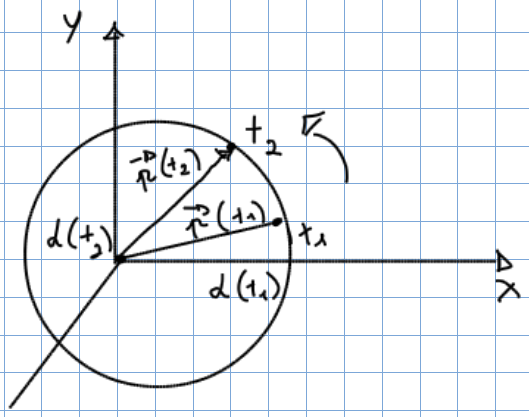
$y(t) = y_{MAX} - \frac{1}{2} g t^2 = 0$

$\Rightarrow \frac{2 y_{MAX}}{g} - t^2 = 0 \Rightarrow t = \sqrt{\frac{2 y_{MAX}}{g}} = \sqrt{\left(\frac{v_0 \sin\alpha}{g}\right)^2 + \frac{2 h_0}{g}}$

Moto circolare uniforme

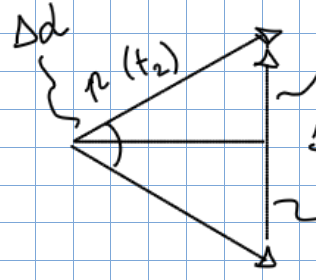
$|\vec{r}(t_1)| = |\vec{r}(t_2)|$ $\quad \vec{r}(t_2) \neq \vec{r}(t_1)$

$\alpha(t_2) > \alpha(t_1)$ $\quad \alpha(t_2) \neq \alpha(t_1)$



$\lim_{\Delta t \to 0} \frac{\Delta\alpha}{\Delta t} = \lim_{t_2 \to t_1} \frac{\alpha(t_2) - \alpha(t_1)}{t_2 - t_1} = \frac{d\alpha}{dt} = \omega$

- $v = \omega R$
- $\vec{v} \perp \vec{r}$
- $a = \omega \cdot v$
- $\vec{a}$ antiparallela a $\vec{r}$ $\quad \vec{a} \perp \vec{v}$



$$\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$$

$$\lim_{x \to 0} \frac{\sin x}{x} = 1 \Rightarrow \sin x \overset{x \to 0}{\approx} x$$

$$\frac{\Delta r}{2} = R \sin\left(\frac{\Delta \alpha}{2}\right) = R \sin\left(\frac{\omega \Delta t}{2}\right)$$

$$R \frac{\omega \Delta t}{2} \Rightarrow \Delta r = \omega R \Delta t$$

$$\frac{\Delta \alpha}{\Delta t} \overset{\Delta t \to 0}{=} \omega \Rightarrow \Delta \alpha = \omega \Delta t$$

$$\Downarrow$$

$$v = \omega R$$

$$|\vec{r}(t_2)| = |\vec{r}(t_1) + \Delta \vec{r}|$$

$$\Delta \vec{r} = \vec{r}(t_2) - \vec{r}(t_1)$$

$$R^2 = \underbrace{|\vec{r}(t_1)|}_{R^2} + |\Delta \vec{r}|^2 + 2 \vec{r}(t_1) \cdot \Delta \vec{r}$$

$$0 = |\Delta \vec{r}|^2 + 2 \vec{r} \Delta \vec{r} \Rightarrow 0 = \frac{|\Delta \vec{p}|^2}{\Delta t} + 2 \vec{r} \cdot \vec{v}$$

$$\Rightarrow 0 = \frac{(\omega R)^2 (\Delta t)^2}{\Delta t} + 2 \vec{v} \cdot \vec{r}$$

$$\Rightarrow 0 = (\omega R)^2 \Delta t + 2 \vec{v} \cdot R$$

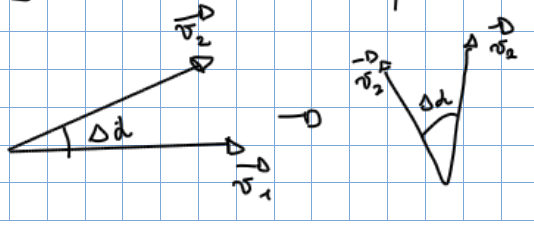
$$\Delta t \to 0$$

$$\Rightarrow 2 \vec{v} \cdot \vec{r} = 0 \Rightarrow \vec{v} \cdot \vec{r} = 0 \Rightarrow \vec{v} \perp \vec{r}$$

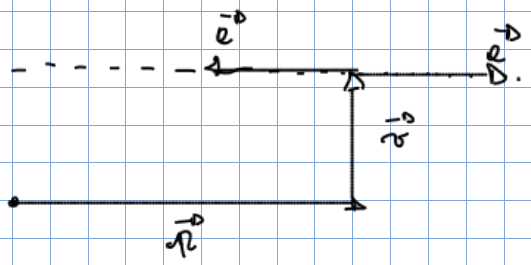
$$\frac{\Delta R}{2} = R \sin\left(\frac{\Delta \alpha}{2}\right)$$

$$\Delta \frac{v}{2} = v \sin\left(\frac{\Delta \alpha}{2}\right) = 2v \sin\left(\frac{\omega \Delta t}{2}\right) \approx v \omega \Delta t$$

$$\Rightarrow a = \frac{\Delta v}{\Delta t} = v \omega$$



$|\vec{r}_2(t_2)|^2 = |\vec{r}_1(t_1) + \vec{r}_{12}|^2$



$\alpha = \angle(\vec{r}_1, \vec{e}) = 0$ $\Rightarrow \vec{r}_1 \cdot \vec{e} = R\alpha > 0$

$\alpha = \angle(\vec{r}_1, \vec{e}) = \pi$ $\Rightarrow \vec{r}_1 \cdot \vec{e} = R\alpha\cos\pi = -R\alpha < 0$

$0 = \vec{r}_2 \cdot \vec{r}_2 \Rightarrow \frac{d}{dt}(\vec{r}_2 \cdot \vec{r}_2) = 0$

$\Rightarrow \frac{d\vec{r}_2}{dt} \cdot \vec{r}_2 + \vec{r}_2 \cdot \frac{d\vec{r}_2}{dt} = 0$

$\Rightarrow \vec{r}_2 \cdot \vec{e} = -\dot{r}^2 < 0 \Rightarrow \vec{r}_2 \cdot \vec{e} < 0$

$\vec{v}$ tangenziale